Good and bad errors

DNA is a highly stable molecule, but sometimes mistakes, known as mutations, occur. These can be in the form of an error, duplication, or omission in the order of the nucleotides A, C, G, and T. Mutation can be spontaneous—the result of errors that occur when the DNA is copied—or may be induced by external influences such as exposure to radiation or cancer-causing chemicals.

Some mutations have no effect, but others may change what the gene produces or inhibit the functioning of a gene. This can lead to problems in the organism as a whole. Examples of disorders caused by gene mutations include cystic fibrosis and sickle-cell disease. Although many mutations are harmful, occasionally a mutation will confer an advantage on an individual, enabling it to survive in its environment better than others of the same species. This type of mutation may end up being passed on through the process of natural selection. Over many generations, mutation is a mechanism for diversification, survival of the fittest, and ultimately evolution.